Summarising and showing data fairly

How do we describe thousands of records in one sentence and one chart without misleading anyone?

AI and Data Science Professional | Session 3 of 24 | Module A: Think with data
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Choose between the mean, median and mode, and say why, using quartiles, the IQR and the standard deviation

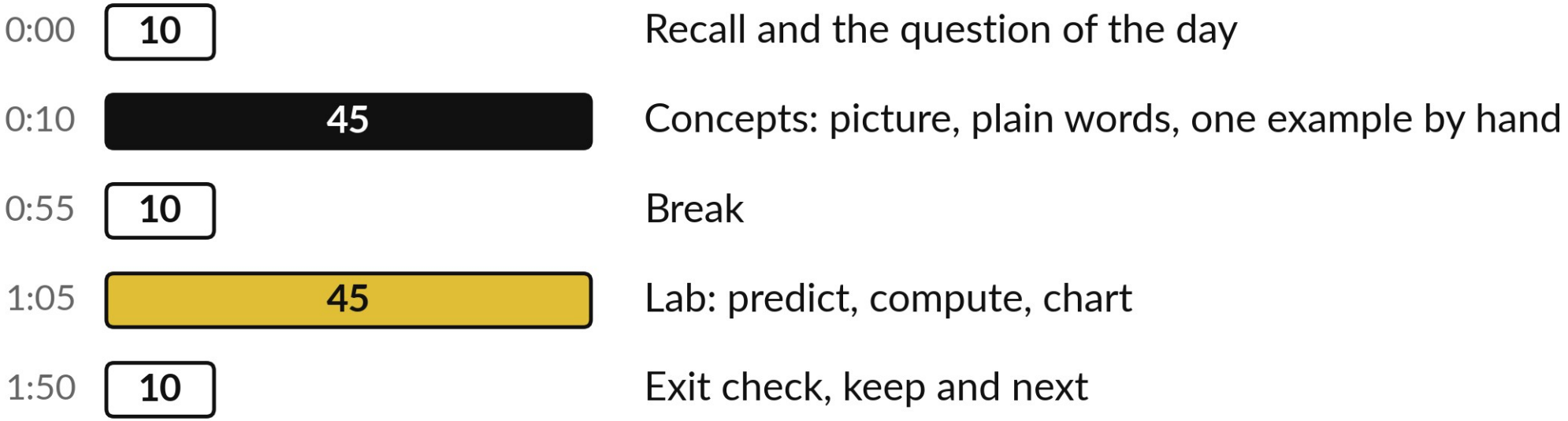


Read a histogram, a box plot and a line chart, and describe a distribution by its skewness



Build a pivot table (a GROUP BY) in Google Sheets, choose a chart for a question, and spot a misleading axis

Your two hours: concepts first, then a Google Sheets lab



Ideas with pictures first. Then you open the sheet. Then the checks and what you keep.

Three questions from last time

1. Order O000003 has 2 products. How many records does it make in orders, and how many in order_lines?

2. Which variable is the primary key of products, and which tab holds it as a foreign key?

3. customer_id is blank on 580 orders. What do you do first?



Answer aloud first. Then we check together.

Answers to the three questions from Session 2

Question	Short answer
Order O000003 has 2 products. How many records does it make in orders, and how many in order_lines?	One record in orders and 2 in order_lines. The two tabs have different granularity.
Which variable is the primary key of products, and which tab holds it as a foreign key?	product_id. The order_lines tab holds product_id as a foreign key.
customer_id is blank on 580 orders. What do you do first?	Read the data dictionary: the blank means a walk-in. Keep the records and add a flag, instead of deleting or filling them.

Words we use today

	Word	What it means
	**Mean, median and mode**	Three ways to say what is typical. The mean is the sum divided by the count. The median is the middle value. The mode is the most common value.
	**Quartiles and IQR**	Quartiles cut sorted values into four equal groups. The interquartile range (IQR) is Q3 minus Q1: the spread of the middle half.
	**Standard deviation**	A typical distance from the mean: the square root of the average squared distance (n minus 1 for a sample).
	**Skewness**	How lopsided a distribution is. Right-skewed means a long tail of large values on the right.
	**Pivot table (GROUP BY)**	A table that groups records and summarises each group with an aggregation such as SUM, COUNT or AVERAGE (MEDIAN in Sheets and pandas). In SQL and pandas it is GROUP BY.

Seven summary jobs on order_value_ugx, each one formula in Sheets

Statistic	Sheets formula	Result
Count	=COUNT(orders!G2:G2001)	2,000
Sum	=SUM(orders!G2:G2001)	99,861,360
Mean	=AVERAGE(orders!G2:G2001)	49,930.68
Median	=MEDIAN(orders!G2:G2001)	24,800
Mode	=MODE(orders!G2:G2001)	7,000 (80 orders)
Minimum	=MIN(orders!G2:G2001)	1,140
Maximum	=MAX(orders!G2:G2001)	1,935,520

All amounts are in UGX. Column G of the orders tab holds order_value_ugx.

The workbook is a sample of the full file, so its figures are close but not identical



Workbook: 2,000 orders

Mean 49,931 UGX

Median 24,800 UGX

Mean is 2.01 times the median



Full file: 30,229 orders

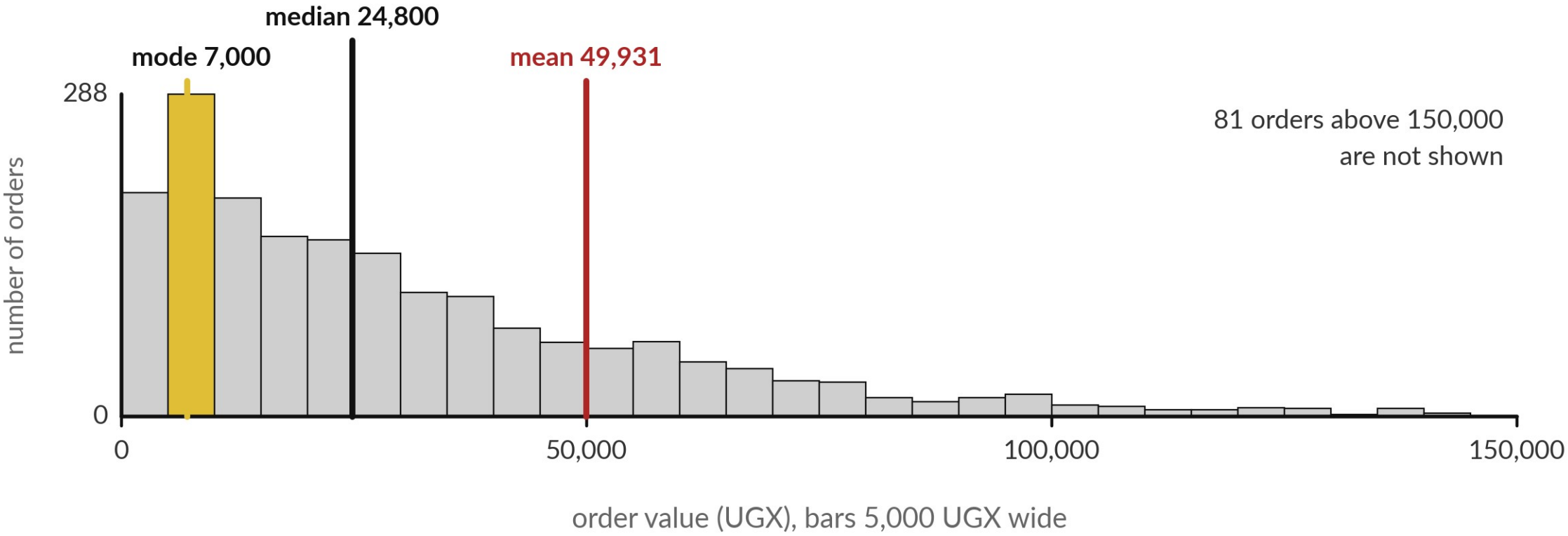
Mean 47,491 UGX

Median 23,200 UGX

Mean is 2.05 times the median

The story is the same. Session 4 explains why the figures differ.

A histogram shows the distribution, and a few large orders pull the mean above the median



Each bar counts the orders in a band. The gold bar holds the mode. The largest order is off the edge.

The mean is the sum of the values divided by how many there are

Mean

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{99{,}861{,}360}{2{,}000} = 49{,}930.68$$

Symbol	What it means	In our data
x-bar	The mean (arithmetic mean)	49,930.68 UGX
n	The number of records	2,000
x_i	The value of record i	order_value_ugx of one order
Σ (sigma)	Add up the values from i = 1 to n	Sum = 99,861,360

In Sheets: =AVERAGE(orders!G2:G2001). The median is the middle value after sorting.

One huge order moves the mean but not the median



Mean

All orders: 49,931 UGX

Without the largest: 48,987 UGX

It moved by about 943 UGX (49,930.68 to 48,987.41)



Median

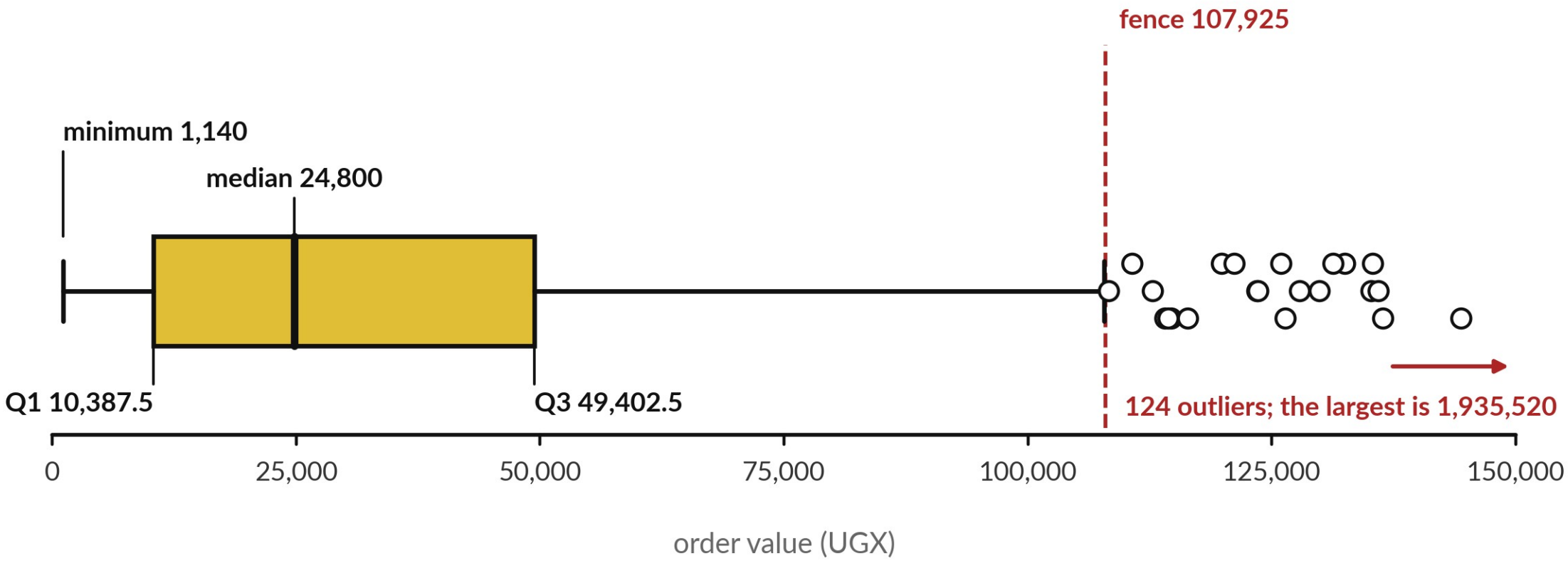
All orders: 24,800 UGX

Without the largest: 24,800 UGX

It did not move at all

For money amounts, give the median beside the mean, and say how many records you counted.

A box plot shows the quartiles: the box holds the middle half of the orders



IQR = Q3 minus Q1 = 39,015 UGX. Beyond Q3 + 1.5 × IQR, the dots are outliers by the IQR rule.

The five-number summary and the outlier fence, each one formula in Sheets

What you want	Sheets formula	Result
Minimum	=MIN(orders!G2:G2001)	1,140
First quartile, Q1	=QUARTILE(orders!G2:G2001,1)	10,387.5
Median, Q2	=MEDIAN(orders!G2:G2001)	24,800
Third quartile, Q3	=QUARTILE(orders!G2:G2001,3)	49,402.5
Maximum	=MAX(orders!G2:G2001)	1,935,520
IQR	Q3 minus Q1	39,015
Orders beyond the fence	=COUNTIF(orders!G2:G2001,">107925")	124

The fence is Q3 + 1.5 × IQR = 49,402.5 + 1.5 × 39,015.0 = 107,925.0. The fence is exactly 107,925, so the formula uses 107925. A percentile such as the 90th is =PERCENTILE(range,0.9).

The standard deviation is a typical distance of the values from the mean

Sample standard deviation

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Symbol	What it means	In our data
s	The sample standard deviation: root of the average squared distance, n − 1 for a sample. Its square is the variance.	128,094 UGX
x_i − x-bar	How far record i sits from the mean (a deviation)	One order minus 49,930.68
(…)²	Square each deviation, so negatives do not cancel	Units are UGX squared
n − 1	The count minus one (ddof=1 in pandas)	1,999

In Sheets: =STDEV(orders!G2:G2001). For a whole population, =STDEV.P divides by n instead.

One example by hand: the standard deviation of five orders

	Order value x (UGX)	Deviation x − mean	Squared deviation
🧾	**66,200**	24,634	606,833,956
🧾	**3,100**	-38,466	1,479,633,156
🧾	**52,200**	10,634	113,081,956
🧾	**79,830**	38,264	1,464,133,696
🧾	**6,500**	-35,066	1,229,624,356
🧮	**Mean 41,566**	Sum 0	Sum 4,893,307,120

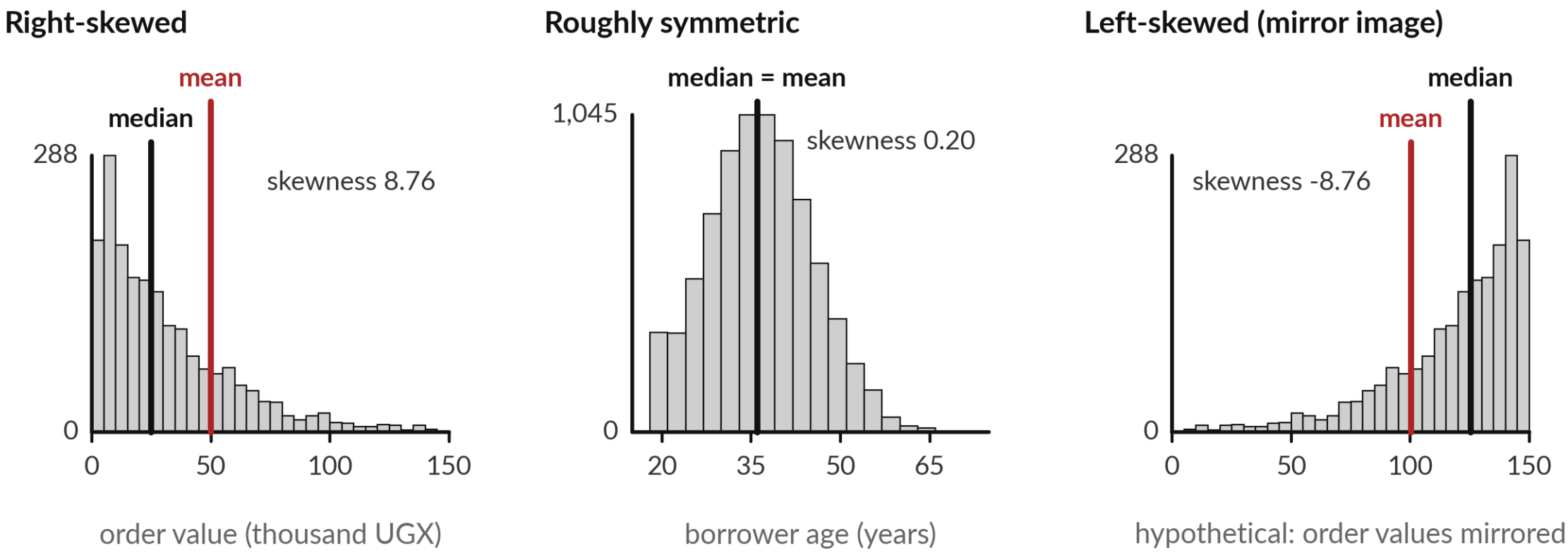
Variance = 4,893,307,120 ÷ 4 = 1,223,326,780. Standard deviation = square root = 34,976 UGX. These are the first five orders in the orders tab.

One large order moves the standard deviation by 7,115 UGX and the IQR by 10

Measure	All 2,000 orders	Without the largest order
Mean	49,931	48,987
Median	24,800	24,800
Standard deviation	128,094	120,979
IQR	39,015	39,005

All amounts are in UGX. The mean and the standard deviation react to extreme values. The median and the IQR resist them.

Skewness describes the shape: a long tail on one side pulls the mean towards it



Left: order values. Middle: ages of 7,998 borrowers. Right: the left picture reversed (hypothetical).

A pivot table is a GROUP BY: it groups the records and summarises each group

Records in the orders tab

order_id	branch_id	order_value_ugx
O000003	B01	66,200
O000046	B01	3,100
O006353	B02	55,800
O006388	B02	27,200

… 1,996 more records

GROUP BY branch_id

Aggregation: COUNT, SUM, MEDIAN

Result: one record per branch

branch_id	orders	sum (UGX)	median (UGX)
B01	372	18,458,585	24,550
B02	295	11,395,000	22,500
B03	232	11,863,530	25,342
B04	256	18,409,915	27,800
B05	202	8,726,725	22,650
B06	256	11,172,340	27,035
B07	182	9,655,610	23,400
B08	205	10,179,655	26,940

Each group is one branch. An aggregation turns many records into one number.

The same GROUP BY question in three tools

	Tool	How you write it	What it returns
	**Google Sheets**	Insert, Pivot table. Rows: branch_id. Values: SUM of order_value_ugx	One row for each branch
	**SQL**	SELECT branch_id, SUM(order_value_ugx) FROM orders GROUP BY branch_id	The same, from a database
	**pandas**	orders.groupby('branch_id').order_value_ugx.sum()	The same, in Python

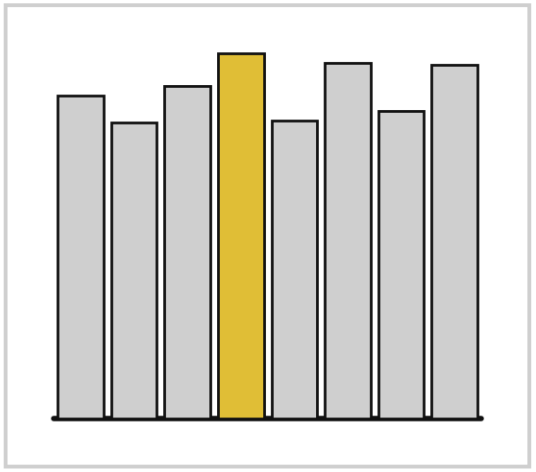
The result is the same in every tool. We meet SQL in Session 10 and pandas groupby in Session 9.

Jinja leads on the mean and the median, but the medians of all eight branches are close

Branch	Orders	Mean (UGX)	Median (UGX)
B01 Kampala Central	372	49,620	24,550
B02 Kampala Ntinda	295	38,627	22,500
B03 Entebbe	232	51,136	25,342
B04 Jinja	256	71,914	27,800
B05 Mbale	202	43,202	22,650
B06 Mbarara	256	43,642	27,035
B07 Fort Portal	182	53,053	23,400
B08 Gulu	205	49,657	26,940

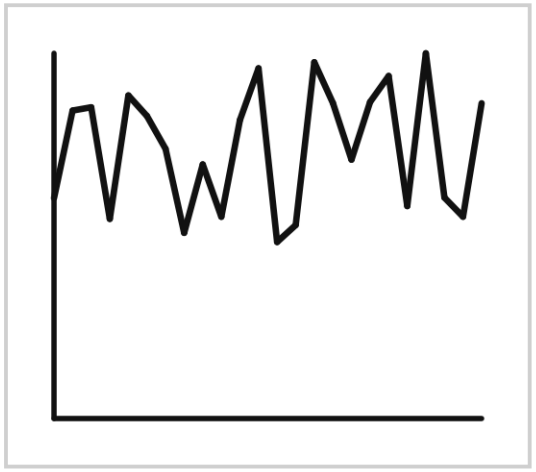
Medians run from 22,500 to 27,800 UGX. Without its largest order (1,935,520 UGX), Jinja's mean falls from 71,914 to 64,605 and Jinja still leads.

Choose the chart from the question you are asking



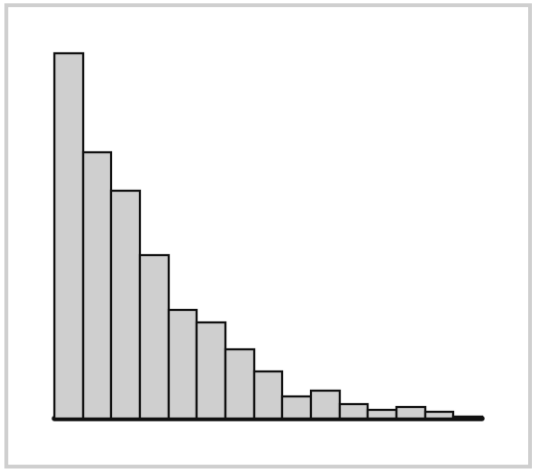
Compare groups

Bar chart: median order for each branch



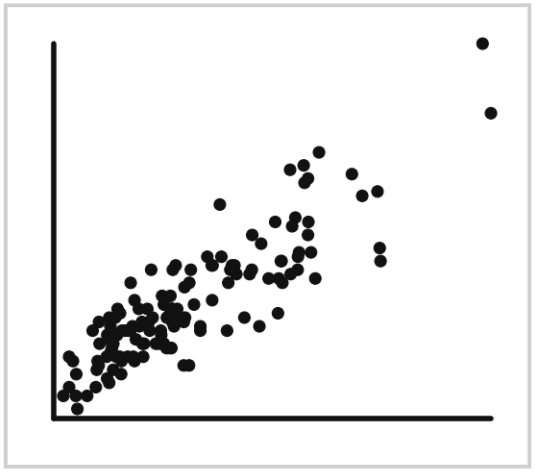
Change over time

Line chart: sales for each month



Shape of one variable

Histogram or box plot: order value

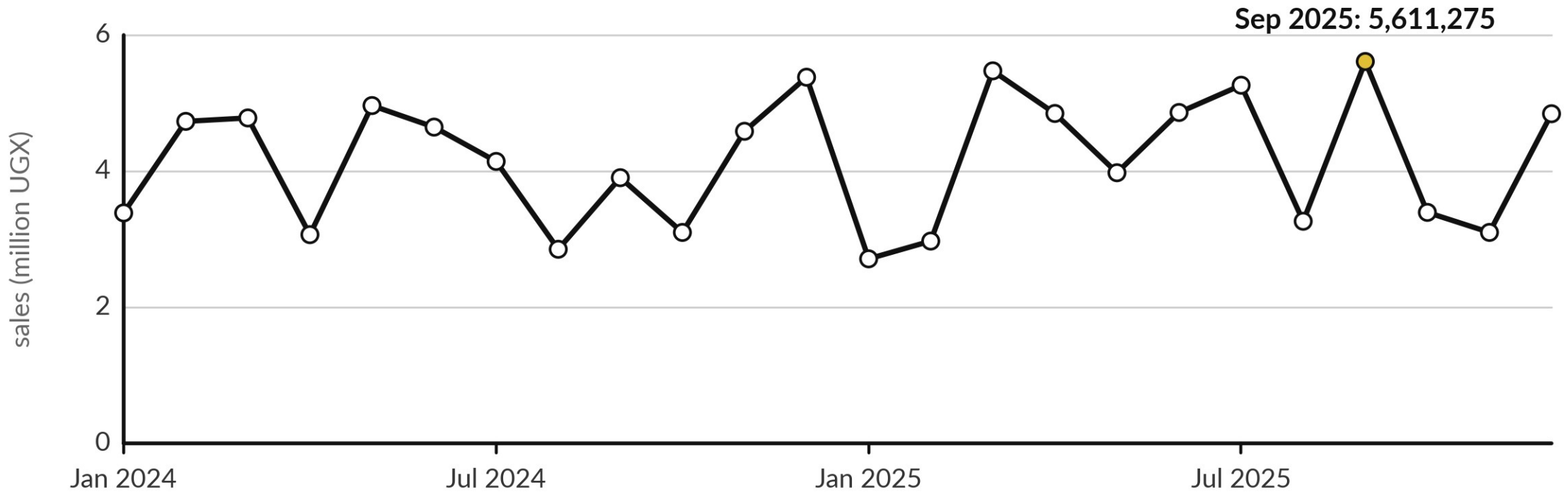


Two variables

Scatter plot: footfall and orders

Real Lakeside data in every panel. Avoid pie charts: people compare angles badly.

A line chart shows change over time; start its axis at zero unless you say clearly that you did not

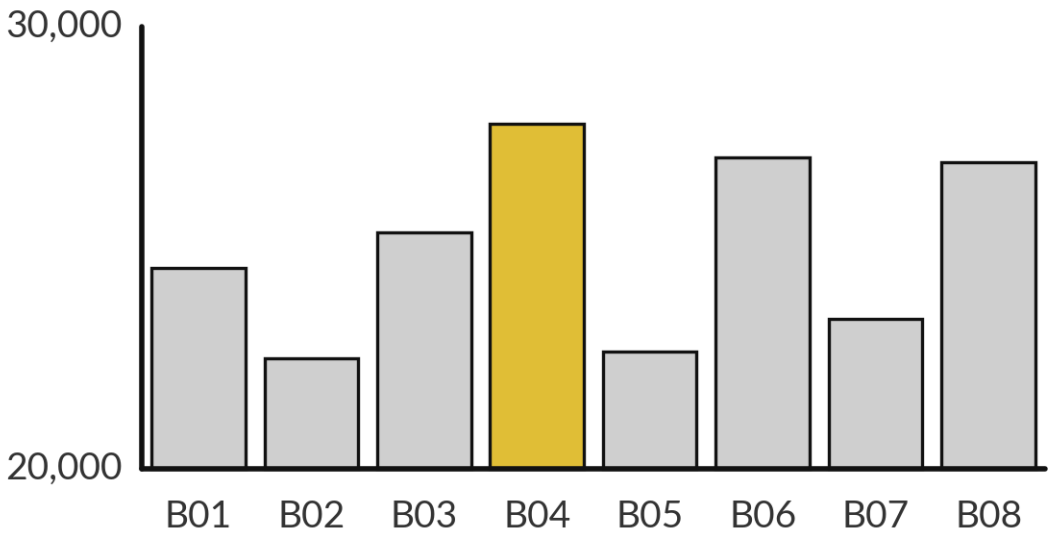


Monthly total of order_value_ugx. About 83 orders a month, so a few large orders can move a month.

A truncated axis makes a 24 percent gap look 3.1 times taller

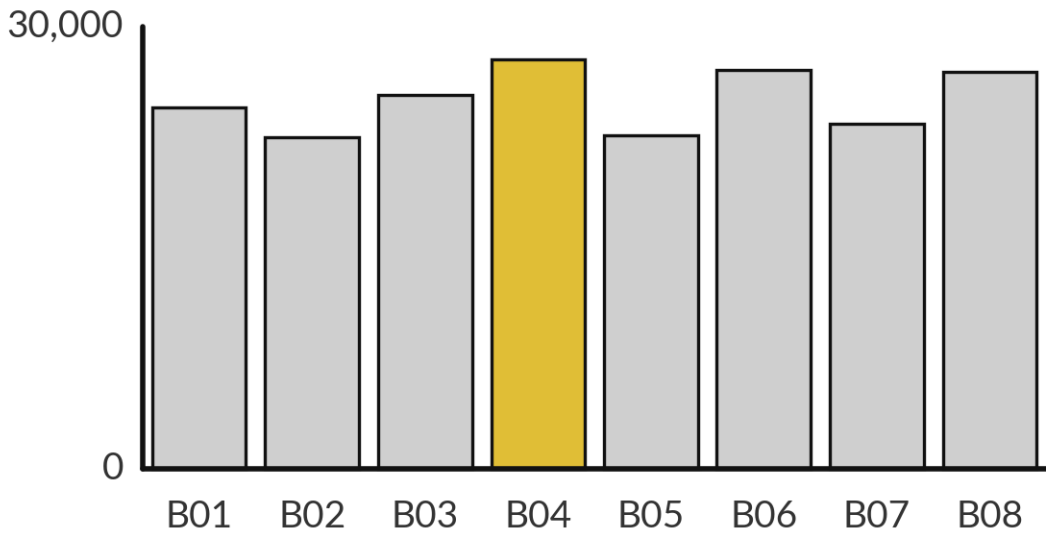
Axis starts at 20,000

B04 looks 3.1 times as tall as B02.



Axis starts at 0

B04 is 1.24 times B02.



The same eight branch medians of order value in UGX. A bar chart must start at zero.

Six checks before you show a chart



Does the title state the point?



Do the axes have labels and units, such as UGX?



Does a bar chart start at zero?



Is the chart type right for the question?



Does it say how many records it shows, and the source?



Would it still look fair to someone who disagrees with you?

Where you will use this at work: report the typical value and the spread beside the total

	Your job	The question	What today lets you do
	**Branch manager at a retail chain, using Excel or Sheets**	What does a typical customer spend?	Report the median order with the IQR and the number of orders.
	**Credit analyst at a bank, using SQL**	How large is a typical loan in each district?	Write a GROUP BY with COUNT and AVG for each district, and take the median in pandas.
	**Health records officer at a district office, using pandas**	Monthly visits for 12 districts	Report the median visits and the IQR for each district before you rank districts.
	**Programme officer at an NGO, handing over to Power BI**	Which region gained most?	Pick a chart from the question, and start bar charts at zero.

Read the recipe before the lab: five formulas summarise a variable

```
=MEDIAN(orders!G2:G2001)
=QUARTILE(orders!G2:G2001,3)
=STDEV(orders!G2:G2001)
=COUNTIF(orders!G2:G2001,">107925")
=LEFT(orders!B2,7)
```

In plain words

Line 1 gives the median: 24,800.

Line 2 gives Q3: 49,402.5.

Line 3 gives the sample standard deviation: about 128,094.

Line 4 counts the outliers beyond the fence: 124.

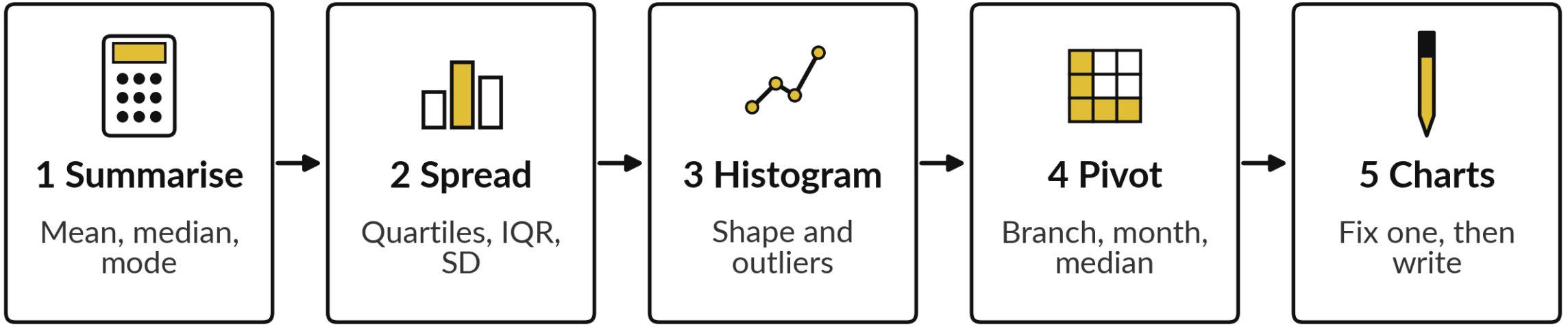
Line 5 takes the first 7 characters of the text date, such as 2024-01. It makes a month column for the pivot table.

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Part 2: the Sheets lab

Close the slides. Open the sheet.

The lab has five steps, and you keep a workbook with pivots and charts



Eight tasks in 45 minutes. For each task, predict the answer, compute it, then write it in a notes tab.

Check your numbers before you move on

What you compute	You should get
Mean, median and mode of order_value_ugx	49,930.68, 24,800 and 7,000
Q1, Q3 and the IQR	10,387.5, 49,402.5 and 39,015
Sample standard deviation, and orders beyond 107,925	128,094 and 124
Orders at most 10,000, at most 50,000 and above 100,000	491, 1,512 and 139
Orders in December 2024	116
Pivot grand total, and December 2024 total	99,861,360 and 5,380,425
Highest and lowest branch median	27,800 (B04) and 22,500 (B02)
Highest and lowest month	Sep 2025 and Jan 2025

If your number differs, find the task that went wrong. Do not change a number to match.

Five likely mistakes in the lab and how to fix them

	What you see	What it means	What to do
✗	**A pivot table shows Count where you wanted Sum**	The summarise setting is COUNTA.	In the pivot editor, set Summarize by to SUM or MEDIAN.
✗	**#NAME? or a blank month column**	The equals sign is missing, or the column is not filled down.	Type =LEFT(B2,7) and drag the fill handle to row 2001.
✗	**A histogram with one tall bar**	The huge orders stretch the axis.	Customize: outlier percentile 95, bucket size 10000.
✗	**A column chart whose axis starts at 20,000**	The axis minimum is set above zero.	Customize, Vertical axis, set Min to 0.
✓	**A bar differs by a few orders from the answer**	Tools treat a bucket edge differently.	Accept a difference of up to 3 orders.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**A manager asks what a typical order is worth. Which statistic do you lead with, and what else do you give?**	The median, 24,800 UGX, with the number of orders and the IQR. The mean, 49,931, is pulled up by a few large orders.
	**What is a pivot table in SQL and in pandas?**	A GROUP BY in SQL and groupby in pandas: group the records, then aggregate each group.
	**A bar chart makes a small gap between two branches look huge. What do you check first?**	Whether the vertical axis starts at zero. Cutting the base makes small gaps look large.

Save your workbook with its pivot tables and charts

What you keep from today

Keep your workbook with three pivot tables, three charts you could show a manager, and two sentences that describe the typical order and the branches.

Our question today

How do we describe thousands of records in one sentence and one chart without misleading anyone?

Next session

Session 4: Uncertainty, samples and cause. We ask how far a sample can sit from the truth, and when a pattern means that one thing caused another.

Take away

Your lab file
A workbook with pivots and charts

Your finding
Two sentences for a manager

Next session
Uncertainty, samples and cause